Knowledge Organiser — Discover: Active Transport

LAUNCH · Edexcel GCSE Biology 1BI0 Foundation · Paper 1 · Topic 1 · Week 6

Key word	What it means
active transport	Movement of substances against the concentration gradient, using energy.
against the gradient	From lower concentration to higher concentration.
carrier protein	A protein in the membrane that moves the substance across.
respiration	The reaction that releases the energy active transport needs.
mitochondria	Where respiration happens; cells doing lots of active transport have many.

Key facts

- Active transport moves substances from LOW to HIGH concentration — against the gradient.
- It requires energy released by respiration.
- It uses carrier proteins in the cell membrane.
- Root hair cells absorb mineral ions from dilute soil water by active transport.
- Cells specialised for active transport contain many mitochondria.

Success criteria

- I can state that active transport moves substances AGAINST the concentration gradient.
- I can state that it requires energy from respiration.
- I can compare all three transport processes.

Discover: Active Transport

Name: ______________________ **Tier ringed by adult:** ◆ Supported ▲ Standard ★ Stretch **Date:** 02/09/2026

Learning objective

I can explain active transport and distinguish it from diffusion and osmosis.

Today at a glance

- Against the gradient
- Energy from respiration
- Compare all three

Today at a Glance

1. Against the gradient
2. Energy from respiration
3. Compare all three

Arrival task

Write one difference between diffusion and active transport.

____

____

We Do — Sort each card into the right group

Active transport needs energy → ________________________________

Diffusion needs energy → ________________________________

Active transport moves substances up the gradient → ________________________________

Active transport is just faster diffusion → ________________________________

Cells doing active transport have many mitochondria → ________________________________

Osmosis moves any particle → ________________________________

WAGOLL

Root hair cells absorb mineral ions by active transport because the concentration of ions is already higher inside the cell than in the soil. Moving them against the concentration gradient requires energy released by respiration, which is why root hair cells contain many mitochondria.

Scaffolding — ▲ Standard

Sentence stems

You give the answer shape for the classic question: 'Mineral ions move into the root hair cell by active transport because their concentration is HIGHER inside the cell than in the soil water. Moving against the gradient requires energy, released by respiration in the mitochondria.' Direction, then why energy is needed, then source.

Word bank

active transport, against the gradient, carrier protein, respiration, mitochondria

Independent Work — ▲ Standard

Task: Complete the comparison table unaided, label the root hair cell, and answer two application questions.

Exit Ticket

Why does active transport need energy?

Assessor Witness Statement

Pearson Edexcel GCSE (9–1) Biology 1BI0 (Foundation, grades 1–5); GCSE Combined Science 1SC0 F where appropriate; AQA UAS science units

Records what was observed, and at which level of independence.

Pupil	____________________
Lesson	Discover: Active Transport — LAUNCH Science · Week 6
Date	____________________

Observed in this lesson

☐ Stated that active transport works against the concentration gradient.

☐ Linked the energy requirement to respiration.

☐ Compared diffusion, osmosis and active transport correctly.

Assessor comment

Ring the tier the pupil actually worked at

◆ Supported	▲ Standard	★ Stretch

Assessor name ______________ Role ____________ Signature ______________ Date ________

Lundy Loop

Space: Three processes at once is a genuine memory load. The comparison table stays visible all lesson — that is deliberate.

Voice: If you can defend a different answer on the sorting task, defend it. Some of these are genuinely close.

Audience: Your comparison table goes into the Topic 1 folder and is the core revision tool at W7.

Influence: The class decides which of the three processes gets extra practice next lesson.

Closing the loop: this one ends on paper — your own hand, or your words scribed by an adult. Saying it aloud starts the loop; writing it is what closes it. Pass is always allowed.

What I said, and what it changed:

Discover: Active Transport — Feedback Sheet

Keep with your portfolio.

Pupil name:	**Date:**
WWW (What Went Well):	
EBI (Even Better If):	
Level worked at: Supported / Standard / Stretch	**Evidence filed:** Yes / Not yet
Pupil response — my next step:	